

Calculus I Lesson 44

$$1) \ a^x = e^{\ln a^x} = e^{x \cdot \ln a}$$

$$D_x(a^x) = D_x(e^{x \cdot \ln a}) = \underline{\hspace{2cm}}?$$

Hint: Let $u = x \cdot \ln a$ and use the Chain Rule $D_x(e^u) = e^u \cdot \frac{du}{dx}$

2) Find $D_x(14^{2x}) = \underline{\hspace{2cm}}?$

Hint: Let $u = 2x$ and use the Chain Rule $D_x(a^u) = a^u \cdot \ln a \cdot \frac{du}{dx}$

3) Find $D_x(4^{-4x}) = \underline{\hspace{2cm}} ?$

Hint: Let $u = -4x$ and use the Chain Rule $D_x(a^u) = a^u \cdot \ln a \cdot \frac{du}{dx}$

4) Find $D_x(x \cdot 3^x) = \underline{\hspace{1cm}} ?$

a) $D_x(3^x) = \underline{\hspace{1cm}} ?$

b) $D_x(x \cdot 3^x) = x \cdot D_x(3^x) + 3^x \cdot D_x(x) = \underline{\hspace{1cm}} ?$

5) Let $f(x) = 4^{-x}$

Find the equation of the tangent line at $(0, 1)$

Hint: $D_x(a^u) = a^u \cdot \ln a \cdot \frac{du}{dx}$

a) $f'(x) = \underline{\hspace{2cm}}?$

b) Slope of tangent line $= f'(0) = \underline{\hspace{2cm}}?$

c) Equation of the tangent is $\underline{\hspace{2cm}}?$

Hint: $y - y_1 = m(x - x_1)$

6) Let $y = x^{3/x}$. Find y' .

Hint: $y = x^{3/x} \Rightarrow \ln y = \ln(x^{3/x}) \Rightarrow \ln y = \frac{3}{x} \cdot \ln x = \frac{3 \ln x}{x}$

Now perform implicit derivative on $\ln y = \frac{3 \ln x}{x}$

For $\ln y$: $D_x(\ln y) = \frac{1}{y} y'$

For $\frac{3 \ln x}{x}$: Use Quotient Rule for Derivative.

a) $D_x\left(\frac{3 \ln x}{x}\right) = \frac{x \cdot D_x(3 \ln x) - 3 \ln x \cdot D_x(x)}{x^2} = \underline{\quad ? \quad}$

b) $y' = \underline{\quad ? \quad}$

7) Let $y = (x - 12)^{x+1}$. Find y' .

Hint: $y = (x - 12)^{x+1} \Rightarrow \ln y = \ln(x - 12)^{x+1} \Rightarrow \ln y = (x + 1) \cdot \ln(x - 12)$

Now perform implicit derivative on $\ln y = (x + 1) \cdot \ln(x - 12)$

For $\ln y$: $D_x(\ln y) = \frac{1}{y} y'$

For $(x + 1) \cdot \ln(x - 12)$: Use Product Rule for Derivative.

a) $D_x[(x + 1) \cdot \ln(x - 12)] = (x + 1) \cdot D_x[\ln(x - 12)] + \ln(x - 12) \cdot D_x(x + 1) = \underline{\hspace{2cm}}?$

b) $y' = \underline{\hspace{2cm}}?$

8) Evaluate $\int 4^{2x} \cdot dx$

Hint: Let $u = 2x$

a) $\frac{du}{dx} = \underline{\hspace{1cm}}?$ and $\frac{1}{2} \cdot du = \underline{\hspace{1cm}}?$

b) Write the integral $\int 4^{2x} \cdot dx$ in terms of u and then evaluate.

$$\int 4^{2x} \cdot dx = \int (\underline{\hspace{1cm}}?) \cdot du = \underline{\hspace{1cm}}?$$

c) $\int 4^{2x} \cdot dx = \underline{\hspace{1cm}}?$

9) Evaluate $\int 4^{2x^2} \cdot x \cdot dx$

Hint: Let $u = 2x^2$

a) $\frac{du}{dx} = \underline{\hspace{1cm}}?$ and $\frac{1}{4} \cdot du = \underline{\hspace{1cm}}?$

b) Write the integral $\int 4^{2x^2} \cdot x \cdot dx$ in terms of u and then evaluate.

$$\int 4^{2x^2} \cdot x \cdot dx = \int (\underline{\hspace{1cm}}?) \cdot du = \underline{\hspace{1cm}}?$$

c) $\int 4^{2x^2} \cdot x \cdot dx = \underline{\hspace{1cm}}?$

10) Evaluate $\int \frac{3^x}{3^x + 4} \cdot dx$

Hint: $\int \frac{3^x}{3^x + 4} \cdot dx = \int \frac{1}{3^x + 4} \cdot 3^x \cdot dx$

Let $u = 3^x + 4$

a) $\frac{du}{dx} = \underline{\hspace{1cm}}?$ and $\frac{1}{\ln 3} \cdot du = \underline{\hspace{1cm}}?$ Hint: $D_x(a^x) = a^x \cdot \ln a$

b) Write the integral $\int \frac{1}{3^x + 4} \cdot 3^x \cdot dx$ in terms of u and then evaluate.

$$\int \frac{1}{3^x + 4} \cdot 3^x \cdot dx = \int (\underline{\hspace{1cm}}?) \cdot du = \underline{\hspace{1cm}}?$$

c) $\int \frac{3^x}{3^x + 4} \cdot dx = \underline{\hspace{1cm}}?$

